

BACE1 inhibitor: next steps that can actually succeed

BACE1_Discovery canvas (print of all tabs)

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Do not run another catalytic occupancy program. Lock a modest, APP-selective product profile, build a substrate-safety assay cascade before chemistry scales, and only then optimize a CNS lead. The program is won or lost on uncoupling sAPP β from sSEZ6 — not on IC₅₀ against a FRET peptide.

Do this first. Any hit that lowers sSEZ6 in parallel with sAPP β is a failed series, even if potent, brain-penetrant, and BACE2-sparing.

Spec	Target
A β / sAPP β reduction	20–40%
sSEZ6 remaining vs vehicle	$\geq 70\%$
APP vs NRG1 / SEZ6 / CatD	$>100\times$
First human cognitive safety read	Week 4–12

1. Target product profile

Attribute	Required	Forbidden (already failed)
Mechanism	APP-selective BACE1 (interface / ASBI). Partial catalytic only as backup if SEZ6 spared at the dose	Orthosteric occupancy aiming at 70–90% BACE1 block
A β PD	20–40% sAPP β and A β _{40/42} (A673T range)	50–90% CSF A β as a success metric
Synaptic safety	sSEZ6 remaining $\geq 70\%$; no 21-day spine loss	sSEZ6 tracking A β one-for-one
Selectivity	BACE1 $\gg$ BACE2, CatD, pepsin, renin; NRG1, CHL1, gp130 spared	BACE2-sparing used as a substitute for substrate selectivity
Molecule	Oral CNS small molecule; unbound brain exposure; not a P-gp substrate	Peptide / TAT conjugate unless a human BBB platform exists
Use	Preclinical amyloid+ prevention, or oral maintenance / add-on after plaque-clearing mAb	Monotherapy in mild-to-moderate AD

Kill the series: neurons with sAPP β down and sSEZ6 or sNRG1 down at the same concentration; 21-day mouse 30% A β dose that drops brain sSEZ6 below $\sim 40\%$ or reduces spines; liver enzymes, retinal CatD, P-gp, or no unbound brain exposure.

Keep the series: APP IC₅₀ ≥ 100 -fold tighter than SEZ6/NRG1/PSGL1; in vivo A β 20–40% with sSEZ6/sCHL1/sgp130/sVCAM-1 unchanged; cognition and two-photon spines like vehicle at ≥ 21 days.

Do not start a virtual screen against the catalytic aspartates until the TPP and selectivity panel exist. Negative controls: verubecestat and a peptide isostere must fail the panel.

2. Assay cascade

Historical BACE1 screens used short peptide FRETs and selected active-site binders. Reward APP engagement; punish SEZ6/NRG1 blockade.

Gate	Assay	Pass	Purpose
0. Controls	Verubecestat, lanabecestat, OM99-2, BACE-IV	They fail the selectivity ratio	Prove the panel can kill known drugs
1. Primary	Membrane APP or MBP-APPC125	Useful IC50; not only a 10-mer FRET	APP-relevant inhibition
2. Selectivity	SEZ6, NRG1, PSGL1; BACE2; CatD	$\geq 100\times$ APP-selective; CatD inactive at 10 μ M	Synaptic and retinal liabilities
3. Mechanism	SPR to APP and BACE1; vs OM99-2	APP and/or exosite; noncompetitive with isostere preferred	Confirm PPI / ASBI
4. Cells	Primary / iPSC neurons	sAPP β and A β down; sSEZ6, sCHL1, sNRG1, sgp130 flat	Physiological membranes
5. CNS	Papp, P-gp, unbound fraction, mouse Kp,uu	Kp,uu ≥ 0.3 ; not an efflux substrate	Avoid peptide BBB failure
6. Safety PD	21-day mouse: A β vs sSEZ6 + two-photon spines	A β -20 to -40% ; sSEZ6 $\geq 70\%$; no spine loss	Program-defining experiment

Approximate remaining vs vehicle (Pratsch 2023): failed Phase 3 both A β and sSEZ6 down together; Shionogi 1 A β 32% / sSEZ6 17% lost spines; Shionogi 2 A β 67% / sSEZ6 39% did not. Target ASBI: A β $\sim 70\%$ remaining, sSEZ6 $\sim 90\%$.

Umibecestat and later BACE1-preferring compounds still caused cognitive/synaptic signals when SEZ6 cleavage fell too far. CSF sVCAM-1 = BACE2 marker; CSF sSEZ6/sSEZ6L = BACE1-synapse marker that decides the dose.

3. Lead to candidate

Months 0–6: CNS-like library on APP-membrane / MBP-APPC125, counterscreen SEZ6 and NRG1 (FAH65 logic). Months 6–18: unbound brain, CatD, hERG, liver, P-gp; co-crystal/HDX to show you are not in the active site. Months 18–30: chronic mouse uncoupling; rat/dog tox (retina, liver); NHP CSF proteomics (A β , sAPP β , SEZ6, SEZ6L, CHL1, gp130, VCAM-1); combo with anti-A β mAb in plaque mice.

In vivo package Phase 3 never used as a gate: 7-day PK/PD; 21-day two-photon (Herms); NOR/Y-maze; plaque mouse $\pm$ mAb; NHP 2–4 week CSF vs verubecestat-like signature.

Chemistry trap. A warhead in the OM99-2 site is a low-dose catalytic backup, not APP-selective.

4. Clinical path — do not rerun EPOCH

Phase	Population	Measure	Go if
1	Healthy, then amyloid+ unimpaired	PK; CSF A β /sAPP β ; sSEZ6, sSEZ6L, CHL1, gp130, VCAM-1; Digit Symbol / RBANS weeks 2-12	20-40% A β , synaptic CSF flat, no umibecestat-pattern worsening
1b combo	Early AD on stable anti-A β mAb	Same CSF + plasma p-tau217 / A β 42/40; ARIA vs mAb alone	Add-on does not worsen cognition or ARIA
2 prevention	Preclinical amyloid+ (AHEAD-like)	CDR-SB / PACC; amyloid PET slope; 12-week cognitive safety	Slower amyloid accumulation, stable cognition, SEZ6-sparing
2 maintenance	Post-mAb plaque clearance	Time to amyloid reaccumulation	Prevents rebound outside Phase 3 occupancy

If resources are limited: (1) TPP + kill criteria; (2) APP vs SEZ6/NGR1 panel with verubecestat fail control; (3) 21-day spines at the 30% A β dose; (4) written use = prevention or mAb-maintenance.

Sources

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